Single-Pinhole Acuity

logMAR Measurement Protocol

Model predictions read from `acuity_sweep_stage4.csv`*,*
and the experiment that tests them

IYPT 2027 Problem 1

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Provenance of every number in this document

Read this first

Every predicted acuity value in this document is read directly from `new_sweeps/acuity_sweep_stage4.csv`, column `logmar`, produced by the `mtf10_vsotf_brightness_adjusted` model. No value here is estimated, interpolated, or assumed. Where a quantity is *not* available from the model — most importantly the bare-eye baseline — this document says so explicitly and does not supply a substitute.

This model failed its one external validation check — read before quoting any number below

The full acuity model (every number in this document) was tested against Wang et al. (2009)'s real clinical pinhole-acuity data, in `stage4_human_vision/RESULTS.md`: $R^2 = -6.39$ to -2.84 across $M = 0, 2, 5$, $p \approx 0.50$. Negative R^2 means worse than guessing the mean; $p \approx 0.50$ means no significant relationship was detected. Every one of the 15 tested points showed the same bias direction — the model over-predicts blur at every point.

The scene-luminance assumption was tested and ruled out as the cause: correcting to Wang 2009's own reported illumination made R^2 *worse*, not better. The project's own leading diagnosis is an uncalibrated formula combined with a small ($n = 5$), non-monotonic clinical dataset — not necessarily wrong optics. Full reasoning in `stage4_human_vision/300TD_KNEE_NOTES.txt`.

This is exactly why the experiment below matters. Every number in this document is a self-consistent, traceable model output — not a validated clinical prediction. The measurement you are about to run is calibration data this model currently has none of.

Chapter 1

What the model predicts, and what it does not

1.1 What it predicts

The model predicts **acuity through a pinhole of stated diameter, at stated myopia**. That is one endpoint of the experiment.

1.2 What it does not predict

The model does **not** predict uncorrected bare-eye acuity. This is not an oversight — it is a direct consequence of the declared scope. Uncorrected acuity in a real eye is set by defocus *plus* higher-order aberrations, accommodation, and neural factors, all of which are explicitly excluded from this project's model. The simulation contains a pinhole in every configuration it can evaluate; there is no aperture-free case in the sweep.

The consequence for your experimental claim

You cannot state a predicted *improvement*. You can state a predicted *endpoint*, and measure the improvement against a *measured* baseline. The defensible sentence is: *"The model predicts logMAR +0.33 through a 400 µm aperture at $M = 8\,D$. We measured logMAR X through that aperture, against a measured bare-eye baseline of Y."*

Anyone claiming a predicted Δ from this model is asserting a bare-eye number the model never produced.

Chapter 2

Predicted acuity for the fabricated designs

Each row pairs the gaze-weighted winning aperture for that myopia level with the model's predicted acuity at exactly that aperture and myopia.

M (D)	Aperture (µm)	MAR (arcmin)	logMAR	Snellen	Retinal illum. (Td)
0	2000	0.45	−0.344	20/9.1	31416
1	1250	0.76	−0.120	20/15.2	12272
2	750	1.19	+0.076	20/23.8	4418
3	750	1.29	+0.112	20/25.9	4418
4	600	1.52	+0.181	20/30.3	2827
5	600	1.74	+0.240	20/34.8	2827
6	500	1.81	+0.259	20/36.3	1963
8	400	2.14	+0.331	20/42.8	1257

Table 2.1: Predicted acuity through each fabricated design. Apertures from `gaze_weighted_winner_per_M.csv`; acuity from `acuity_sweep_stage4.csv`.

An independent consistency check, worth stating at defence

These winning apertures were selected by the *gaze-weighted geometric* evaluator — viability, ghost margin, blind fraction — with no acuity term in its objective. Yet at every myopia level from 0 to 8 D, the selected aperture is **also** the acuity-optimal aperture in the independent Stage-4 sweep. Two evaluators built on different objectives agree on all eight designs. That agreement was not enforced and is not a coincidence worth passing over.

Chapter 3

The strongest testable prediction you have

3.1 Acuity through a fixed small aperture is nearly independent of myopia

Holding the aperture at 400 µm and varying only myopia, the model gives:

M (D)	logMAR	Snellen
0	+0.341	20/43.9
1	+0.337	20/43.4
2	+0.333	20/43.0
3	+0.329	20/42.7
4	+0.327	20/42.4
5	+0.325	20/42.3
6	+0.325	20/42.2
8	+0.331	20/42.8

Table 3.1: Predicted acuity through a fixed 400 µm aperture across the full myopia range. Total spread: 0.016 logMAR.

Across eight dioptres of myopia the predicted acuity varies by **0.016 logMAR** — roughly a twentieth of one chart line, and far below the ±0.1 logMAR repeatability of human acuity testing. At 400 µm the aperture, not the refractive error, sets the acuity.

Why this is the experiment to run

This prediction is **falsifiable, sharp, and requires no knowledge of anyone's exact prescription**. Recruit subjects across a range of myopia. If the model is right, all of them read approximately the same line through the same 400 µm pinhole — while their bare-eye acuity differs enormously.
That is a far stronger experimental design than measuring one subject and comparing to one number, because the *shape* of the result carries the claim. A flat line across subjects of very different refractive error is difficult to produce by accident.

3.2 Why the flatness happens

Two effects nearly cancel. As myopia increases, the defocus blur circle grows — but the pinhole is already narrow enough that the accepted cone is set by the aperture, not by the eye's focusing error. The residual defocus contribution is small compared with the diffraction floor the 400 µm

aperture imposes. The predicted acuity therefore tracks the diffraction limit of the aperture, which does not depend on M at all.

The flatness breaks at larger apertures, where defocus reasserts itself. Compare 2000 µm: logMAR -0.344 at $M = 0$ but $+1.330$ at $M = 8$ — a spread of 1.67 logMAR, a hundred times larger. That contrast is itself a second testable prediction, and a good slide.

Chapter 4

Building the chart

4.1 logMAR, defined

logMAR is $\log_{10}$ of the minimum resolvable angle in arcminutes. A one-arcminute detail gives logMAR $= 0$, conventionally 20/20. Positive is worse, negative is better. One chart line is 0.1 logMAR by construction, so a line's letters are $10^{0.1} = 1.259\times$ the size of the next line down.

$$\text{logMAR} = \log_{10}(\text{MAR in arcmin}), \qquad \text{Snellen denominator} = 20 \times 10^{\text{logMAR}} \tag{4.1}$$

4.2 Letter sizing

A Snellen letter is five times the stroke width, and the stroke subtends the MAR. So at viewing distance D, the letter height is

$$h = 5 \times \text{MAR}_{\text{arcmin}} \times \frac{\pi}{180 \times 60} \times D = 1.4544 \times 10^{-3}\,\text{MAR}_{\text{arcmin}} \times D \tag{4.2}$$

At $D = 4.0\,\text{m}$:

logMAR	MAR (arcmin)	Snellen	Letter height (mm)	Stroke (mm)
+0.7	5.01	20/100	29.2	5.83
+0.6	3.98	20/79.6	23.2	4.63
+0.5	3.16	20/63.2	18.4	3.68
+0.4	2.51	20/50.2	14.6	2.92
+0.3	2.00	20/39.9	11.6	2.32
+0.2	1.58	20/31.7	9.2	1.84
+0.1	1.26	20/25.2	7.3	1.46
0.0	1.00	20/20	5.8	1.16
−0.1	0.79	20/15.9	4.6	0.92
−0.2	0.63	20/12.6	3.7	0.73

Table 4.1: Letter dimensions at 4.0 m viewing distance. Scale linearly for other distances.

Why 4 metres, and why this range

The predictions cluster between logMAR −0.34 and +0.34, so the chart must span at least −0.2 to +0.7 to bracket both the predicted pinhole endpoints and plausible bare-eye baselines for mild myopes. At 4 m the smallest stroke is 0.92 mm — printable on a domestic laser printer. At 2 m it would be 0.46 mm, at the edge of print fidelity, and printer dot structure would start to limit the measurement rather than the eye.

4.3 Construction

Five letters per line, Sloan set (C, D, H, K, N, O, R, S, V, Z — equal legibility by design), even spacing of one letter width. Print black on matte white; glossy paper produces specular glare that confounds the low-transmission conditions. Verify the print scale by measuring a printed reference bar with callipers before use, exactly as with the etch masks.

Chapter 5

The confounder that will ruin this measurement

5.1 Retinal illuminance is not a footnote

The predicted-acuity table carries a Troland column for a reason. At $M = 8$ the 400 µm design delivers 1257 Td against 31416 Td for the 2000 µm case — a factor of 25. The model's `mtf10_vsotf_brightness_adjusted` predictions already include a brightness adjustment, but they assume a specific scene luminance (10 000 cd/m^2, a bright outdoor surface).

If you test indoors, the model's assumption is violated

A typical indoor chart under room lighting is 100–300 cd/m^2, some 30–100× dimmer than assumed. Through a 400 µm aperture at 1% transmission, the retinal illuminance can fall low enough that acuity becomes photon-limited rather than optics-limited — and you will measure a brightness effect while believing you measured a diffraction effect.

Mitigation: illuminate the chart hard. Direct lamps onto the chart to raise its luminance as high as you can and *measure it* with a light meter (a phone app with a calibrated lux reading is adequate to within a factor of two, which is enough to know whether you are in the right regime). Record the value. If you cannot reach high luminance, state that the bare-eye and pinhole conditions were measured at different effective retinal illuminance, and that the comparison is therefore not purely optical.

5.2 Accommodation

At 4 m a myopic eye cannot accommodate to bring the chart into focus — accommodation only adds power, moving focus nearer, which is the wrong direction. That is convenient: it means the defocus you are testing is stable. **Do not test at short distances**, where accommodative effort varies between trials and between conditions.

5.3 Pupil diameter

The model assumes a 4 mm pupil. Under the bright illumination this protocol requires, real pupils constrict to 2–3 mm. A smaller pupil improves bare-eye acuity by itself — a natural pinhole effect — which *reduces* the apparent benefit of the artificial aperture. Photograph the eye against a mm scale under test lighting, or at minimum record the lighting conditions so the direction of this bias is documented.

5.4 Learning effect

Subjects memorise charts. Use at least two different letter arrangements and alternate them, or randomise the letter order between repeats. Otherwise later conditions score artificially well.

5.5 The ranked list of things that will go wrong

1. Chart too dim, so you measure photon noise instead of optics.
2. Chart printed at the wrong scale — verify with callipers.
3. Pupil constricted under bright light, shrinking the measured benefit.
4. Subject memorising the chart.
5. Pinhole not centred on the pupil, so the subject sees through its edge or misses it entirely.
6. d_{gap} varying between conditions because the mount is hand-held.

Chapter 6

Protocol

6.1 Fixed conditions — record once per session

Quantity	Value	How obtained
Viewing distance	_____ m	tape measure, ± 0.02 m
Chart luminance	_____ cd/m^2 or lux	light meter / phone app
Chart version(s) used	_____	A / B alternation
d_{gap} (mount to cornea)	_____ mm	spacer, fixed
Room lighting	_____	description
Date / time	_____	

6.2 Per-subject record

Quantity	Value	Source
Subject ID	_____	anonymised
Refractive error, tested eye	_____ D	optometrist Rx if available
Rx date	_____	
Astigmatism (cyl)	_____ D	note if > 1 D
Eye tested	L / R	occlude the other
Pupil diameter under test light	_____ mm	photo vs. scale, or note

If prescriptions are unavailable

Chapter 3's flatness prediction does not need them. Rank subjects qualitatively ("no glasses", "mild", "strong") and test whether pinhole acuity is flat across that ranking while bare-eye acuity is not. Report refractive error as unmeasured. A qualitative ordering with an honest caveat is worth more than a fabricated dioptre value.

6.3 Procedure

1. Occlude the untested eye. Keep it open under the occluder.

2. Seat the subject at the fixed distance, head steady.
3. Present the chart. Subject reads from the largest line downward, aloud, letter by letter.
4. Score a line as read if at least 3 of 5 letters are correct. The threshold is the smallest such line.
5. Record the line's logMAR.
6. Rest 30 s. Change chart version.
7. Repeat for the next condition in randomised order.

Conditions: (1) bare eye, (2) single 400 µm pinhole, (3) fabricated array. Three repeats each, order randomised across all nine trials.

6.4 Recording table

Trial	Subject	Condition	Chart	Threshold line	logMAR
1					
2					
3					
4					
5					
6					
7					
8					
9					

Chapter 7

Analysis

7.1 Per condition

Mean and standard deviation of the three repeats. Human acuity repeatability is typically ±0.1 logMAR, so a standard deviation much above 0.15 indicates a procedural problem — most often inconsistent pinhole centring — rather than a real effect.

7.2 Test 1 — the endpoint comparison

For each subject, compare measured pinhole acuity against the model's prediction at that aperture and that subject's myopia (Chapter 3 table).

$$\text{residual} = \text{logMAR}_{\text{measured}} - \text{logMAR}_{\text{predicted}} \tag{7.1}$$

Report the residual, signed. A positive residual means the eye did worse than predicted. Interpretation:

- $|\text{residual}| < 0.15$: agreement within measurement repeatability.
- Residual $> +0.3$ consistently: a confounder is active. Check chart luminance first — the model assumes bright outdoor conditions.
- Residual < -0.3: the eye outperforms the model. The model omits neural processing and excludes higher-order aberrations, both of which can favour the real eye.

7.3 Test 2 — the flatness test (the strong one)

Plot measured pinhole logMAR against myopia (or qualitative rank) for all subjects. The model predicts a spread of 0.016 logMAR across 0–8 D — effectively a horizontal line. On the same axes plot bare-eye logMAR, which should fall steeply with increasing myopia.

What each outcome licenses you to say

Pinhole flat, bare-eye steep: the central claim of the whole project, measured. *"Through a 400 µm aperture, acuity is set by the aperture rather than by refractive error — measured across subjects spanning X dioptres."*
Pinhole also falls with myopia: the model is wrong in a specific, interesting way, and the slope tells you how wrong. This is a publishable-grade negative result, not a failure.
Both flat: your subjects are not myopic enough to separate the conditions, or the chart is too easy. Extend the chart to smaller lines and recruit a stronger myope.

7.4 Test 3 — array versus single hole

The sparse array is designed so that only one aperture illuminates the pupil at a time. Two consequences follow, both testable:

1. **On axis, array acuity should equal single-hole acuity** to within repeatability, since the eye is looking through one hole in both cases.
2. **Off axis, the array should keep working and the single hole should not.** Rotate the subject's gaze 10–20° by moving the chart laterally while the mount stays fixed. The single hole should go dark; the array should hand over to a neighbouring aperture.

Also record: **does any subject report a doubled image at any gaze angle?** The model predicts none, at any myopia, because no ghost pair ever forms. An observed ghost would falsify that prediction and is worth more than confirming it.

Chapter 8

Reporting

8.1 The claim, written honestly

"Through a 400 µm aperture at $d_{\text{gap}} =$ _____ mm and _____ m viewing distance, N subjects spanning _____–_____ D of myopia read logMAR _____ ± _____, against bare-eye acuity ranging from _____ to _____. Our model predicts $+0.33 \pm 0.01$ across that same range; this model has not previously been validated against real clinical data, and our one prior validation attempt (against Wang et al. 2009) found no significant relationship ($R^2 < 0$). This measurement is that calibration."

8.2 What must accompany it

- Chart luminance, measured, with a statement of whether it met the model's assumed regime.
- Pupil diameter, measured or noted as unmeasured.
- Sample size and how refractive error was established.
- Every subject's raw data — do not report only the mean.
- Any subject excluded, and why, decided before analysis.

Do not report a predicted improvement

The model produces no bare-eye baseline. Any Δ you quote must have a *measured* baseline on both sides. Quoting a predicted improvement means quoting a bare-eye number the model never generated — which is exactly the kind of unsourced figure this project has otherwise been careful to avoid.